CLASS 9 · CBSE · GANITA MANJARI PART 2 · DPP SHEET

Chapter 11: The World of Algorithms

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Execute the algorithm divisors(12) and write the list.
2. Find $23 \bmod 7$, $100 \bmod 9$ and $45 \bmod 15$.
3. Write the list of common divisors of 18 and 24, and their gcd.
4. Find gcd(35, 14) by Euclid's subtraction algorithm. Show the reductions.
5. Find gcd(180, 48) by the division algorithm.
6. Find gcd(1001, 143).
7. Find gcd(391, 299).
8. For gcd(51, 2), which version of Euclid's algorithm is faster? How many reductions does the division version need?
9. Use the gcd to find the lcm of 12 and 18.
10. Use the list of divisors to decide whether 29 is prime.

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Answer Key

1. [1, 2, 3, 4, 6, 12]
2. 2, 1 and 0
3. [1, 2, 3, 6]; gcd 6
4. $\gcd(35, 14) \to \gcd(14, 21) \to \gcd(21, 14) \to \gcd(14, 7) \to \gcd(7, 7) \to \gcd(7, 0)$; answer 7
5. $180 \bmod 48 = 36$, $48 \bmod 36 = 12$, $36 \bmod 12 = 0$; gcd 12
6. $1001 \bmod 143 = 0$, so the gcd is 143
7. $391 \bmod 299 = 92$, $299 \bmod 92 = 23$, $92 \bmod 23 = 0$; gcd 23
8. The division version; 2 reductions ($\gcd(2, 1)$, then $\gcd(1, 0)$)
9. gcd 6, so lcm $= \dfrac{12 \times 18}{6} = 36$
10. divisors(29) = [1, 29] has exactly two elements, so 29 is prime

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